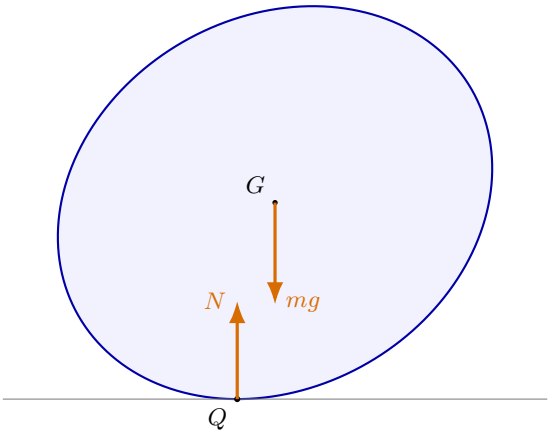
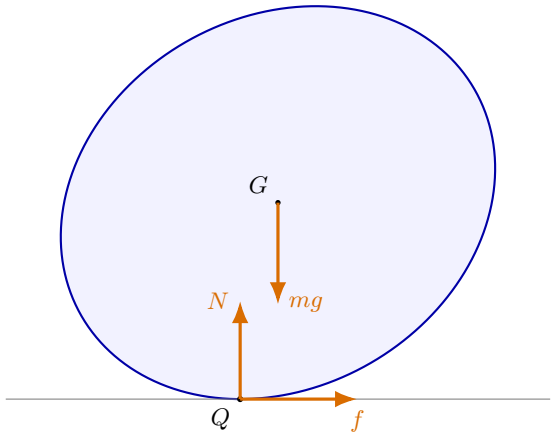
光滑接触：$f = 0$



纯滚动：该时刻 $f > 0$